# Psycho Acoustic Testing to Determine the Optimal Frequency for Audible Safety Alerts for Freediving

Arnav Nanda, Mohammad Zoraiz, Dhruva Barua, Raghad Mohamed

A. N. (phone: 919-600-3361; email: arnav.nanda@duke.edu), M. Z. (phone: 440-771-3643; email: mohammad.zoraiz@duke.edu), D.B. (phone: 703-447-9748; email: dhruva.barua@duke.edu), R.M. (phone: 929-267-9606; email: raghad.mohamed@duke.edu)

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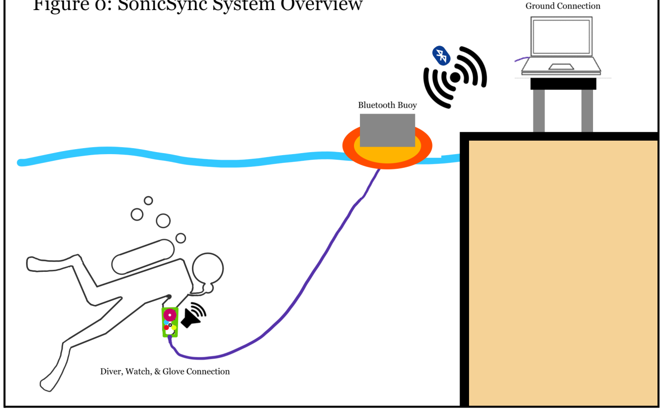
***Abstract* — Freediving is a captivating sport that involves diving underwater on a single breath, pushing the boundaries of human capability and exploring the depths of the ocean with minimal equipment. While free diving offers a unique and thrilling underwater experience, it also poses several difficulties. Having been in the water for long periods, many divers can develop hypoxia or decompression sickness. To solve this issue many divers use watches that allow them to monitor the heart rate, oxygen levels, and depth levels. Most of these watches can track and monitor vital signs and warn in case of potential harm or risk to the diver. While these watches are helpful, they are not preventative. This project aims to address these problems and improve upon current products produced in the free diving industry. Duke's EGR102 team's goal is to create an effective data collection and signal analysis algorithm dedicated for efficiently analyzing data regarding different pitches, timbers, and audible tones at varying depths of water. More importantly, the product should be able to stay underwater at 6 meters depth for more than an hour.**

Figure 0: SonicSync System Overview




## I. INTRODUCTION

This project aims to enhance audible safety alerts for freediving and scuba products, addressing the critical need for improved underwater safety. Current methods face challenges in efficiently generating and evaluating new alerts, contributing to diver casualties due to ineffective tones. We seek to develop a submersible device that enables real-time feedback from divers underwater, improving warnings for dive conditions, neutral buoyancy, target depth, and variometer usage. By reducing underwater risks and enhancing user experience, the project contributes to technological advancements and safety standards for underwater exploration.

Our testing apparatus comprises distinct diver and developer sections. The diver section encompasses the speaker output and rating systems, while the developer section integrates advanced signal analysis applications and diver-to-developer communication channels. In the diver design section, we employ a force sensor-based rating system, enabling divers to provide specific ratings for various tones. There are two force sensors, one in the palm recording specific ratings (very bad (0), bad(1), Average (2), good (3), great(4), best (5)), and the other one on the back of the hand serving as a confirmation button. Adhering to the diver's constraints, the device prioritizes ease of use, portability, and a low form factor. Notably, the tone generator is engineered for durability, submersible up to six feet deep for six hours, and can emit audible frequencies to divers. The developer section facilitates effective communication with divers by enabling the transmission of signals and messages. This communication interface is realized through an application developed using MATLAB app designer, providing developers with higher-level signal analysis and creation capabilities. Data transfer is optimized through a buoy that serves as a conduit for sending and receiving signals and data, facilitating complex data analysis. These design elements empower divers to provide impactful feedback and enable expeditious data analysis on the developer's end.

The team's objective is to establish a system with 98% accuracy in receiving specific diver feedback. This entails the device being capable of emitting frequencies within the 200Hz to 20kHz range at an approximate sound pressure level (SPL) of 70dB, with a permissible variance of $\leq$ 3dB above and below this threshold. To streamline communication between the device and a laptop, a tailored diver protocol leveraging an optimized Bluetooth buoy, targeting a swift ten-second maximum for data transfer in both directions.

The testing approach focuses on creating a resilient design that offers clear and concise diver feedback, specifically assessing the psychoacoustic effectiveness of different tones. Preliminary tests have successfully identified an efficient feedback system utilizing buttons and potentiometers, independently and in conjunction with a Bluetooth buoy. The following paper will provide comprehensive details on the validation and testing outcomes, with a particular emphasis on the details of systems integration and design.

Previous prototypes included using a potentiometer-based design that allows the diver to turn a dial to a specific setting allowing for classification of sound. By turning the potentiometer, there is an increased resistance, lowering the input voltage, and allowing for classification. Ethernet cables were used to streamline the number of wires used to send data to and from the Arduino and computer. Each cable has five individual wires allowing for only one data cable to be used to send instructions to both the speaker and potentiometer.

An alternative prototype consisted of fully enclosed compact laser-cut acrylic box housing buttons, an Arduino, and a speaker integrated into a watch-like device securely fastened onto the user. Utilizing a micro-USB cord as the sole electronic connection to the computer above water, the device facilitated diver feedback via four color-coded buttons and a confirm button for affirming selections. Above water, an audio engineer received real-time wired feedback, generating corresponding tones transmitted underwater, ensuring seamless communication between the diver and engineer.

## II. METHODOLOGY AND CREATION OF DEVICE

After successfully creating a functional prototype, our goal was to enhance and automate our design by incorporating new mechanical and electrical features. These improvements included transitioning from traditional buttons and potentiometers to more efficient data allocation systems. This change not only reduced the number of wires needed but also minimized the requirement for waterproofing our devices. Additionally, we introduced innovative mechanical approaches, such as designing a buoy to house our electronics on the water surface.

Our system comprises three distinct components: the glove, the buoy, and the software. The glove features two integrated sensors, offering divers a low-profile, comfortable experience. The buoy serves as the electronic hub, facilitating communication between developers and divers. Significantly reducing the need for waterproofing by combining all electronics into a single, secure, waterproof enclosure [Figure 0].

### *A. Glove Iteration 1: Flex Sensors*

The first iteration of our glove design included flex sensors as a rating system. Each flex sensor would correspond to a finger of a glove, allowing the diver to assign a specific rating from one through four (worst to best) for a specific tone. By changing the resistance in each flex sensor, i.e. the diver bending their finger, the developer would receive a specific rating. To make sure an ESP32 microcontroller device reads signals, our team created an operation amplifier circuit that allowed for each flex sensor to have a large change in resistance when bent. For example, having the ring bent could notify the developer of a rating corresponding to one for worst indicating a potential issue with the system.

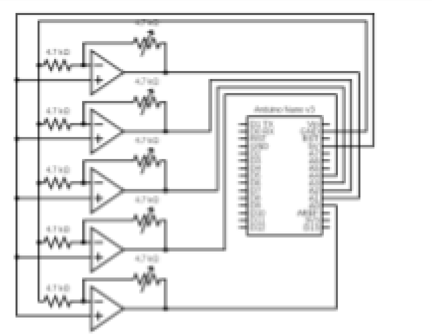


Figure 1: Flex Sensor circuit diagram

As seen in Figure 1, our flex sensor design consisted of a Texas instrument LM741 Operational Amplifier (OP amp), five flex sensors, and an ESP32 microcontroller. By running a 3.3V power source through the amplifier, we wanted to increase the resistive range outputted by the flex sensor. We also wanted to test the difference between different types of flex sensors to determine which brand would be able to provide the largest resistive range. However, a key issue arose with this design iteration— the flex sensors did not output a large enough resistance range. This limitation made it extremely challenging to distinguish between bent and unbent fingers. Despite using an operational amplifier, the flex sensors were unable to exhibit a substantial change in resistance, making data classification difficult.

## B. Glove Iteration 2: Force Sensors

To tackle the problems faced with flex sensors, our team decided to approach another resource of variable resistors—the force sensor. Known as a force transducer, a force sensor can transform an applied mechanical force—including load, weight, tension, compression, or pressure—into an electrical output signal that can be measured, converted, and standardized. Instead of using five flex sensors, we decided to use a single force sensor, allowing the diver to tap multiple times to get specific ratings. Alongside one force sensor to determine specific ratings, we used another force sensor as a confirm button. This allows the diver to determine specific ratings and confirm them. As seen in Figure 2, we only need two wires (ground and ACDC input for nano), which significantly reduces the number of wires used as compared to that of the flex sensors.

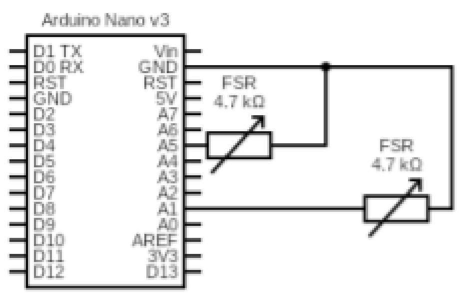



Figure 2: Force Sensor circuit diagram

Once more, we ran a significant number of tests on the force sensors to make sure that they would be able to withstand different environments and wire lengths. The force sensors were able to impede current flow at 6 lengths of wire and longer. It was also able to withstand water and saltwater solutions over a significant time and repeated use.

## C. Glove Speaker Setup & base

To enhance the speaker design, we implemented significant upgrades from the previous prototype, optimizing functionality and user experience. A key enhancement involved eliminating the need for an audio jack connection between the computer and the ESP32 Nano. Instead, leveraging Bluetooth technology, signals are now directly transmitted to the ESP32 (see Section 3: Matlab, Arduino). These signals are subsequently relayed to the speaker unit, as depicted in Figure 3. This elimination of wired connections

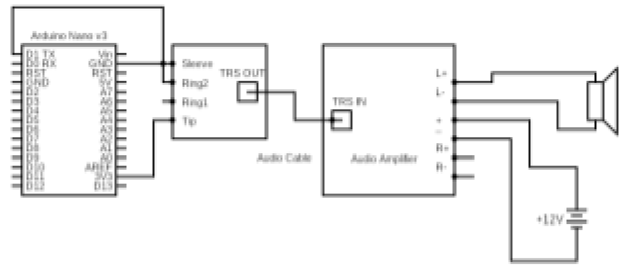



Figure 3: Audio jack setup

not only streamline the system but also enhance waterproofing while consolidating the major electronic components.

Specifically, the TX port of the ESP32 Nano is interfaced with the pinout ring of a TRRS (Tip, Ring 1, Ring 2, and Sleeve) breakout board. The Tip and Sleeve connections supply power and generate a signal, respectively, facilitating seamless communication with the audio amplifier board (see Figure 3). This configuration ensures a robust audio output that is faithfully reproduced by the speaker unit.

We engineered a custom speaker holder using CAD technology to securely mount the speaker on the diver's wrist, complemented by watch bands for a snug fit. As illustrated in Figure 4, this CAD model represents a refined iteration of our previous prototype, characterized by a sleek and low-profile design that seamlessly integrates with the diver's gear, accommodating diverse users with ease.

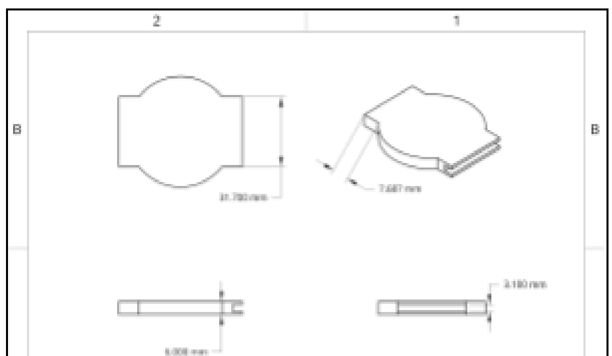


Figure 4: Watch Base

## D. Bluetooth Buoy

To house all of our electronics and minimize waterproofing, we decided to create a Bluetooth buoy - a 3d printed device that would float above the water to give an approximate location of the diver and send data between the device and the diver. The buoy is composed of two different sections: the floating section, and the electrical housing. The electrical housing consists of two boxed sections that can hold our custom-made chip and integrated microcontroller, an audio amplifier in place, while floating, and a small one that will allow for a wire to be fed down to the diver. This wire serves as the main communication device between the diver and the buoy. The second section of the buoy is the floating device - consisting of four wings fitted together and a foam noodle to gain as much surface area to create a buoyant structure (figure 5).

The buoy from Figure 5 was painted red to allow developers and other necessary staff to locate the diver with ease, in case of emergencies. The top lid of the buoy has multiple holes to allow for wires and other electronics to be added on if necessary. Some examples included adding a Bluetooth amplifier, or light to either boost the Bluetooth signal or allow the diver to be more visible.

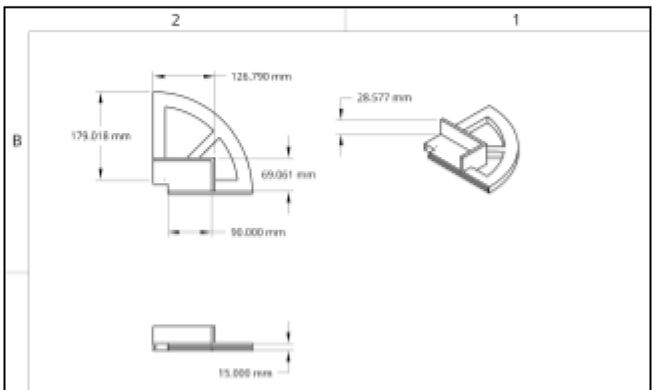



Figure 5: Buoy Design

## III. CODE

Our system includes a seamless integration of software-to-hardware communication facilitated by MATLAB and Arduino IDE. Leveraging Bluetooth connectivity, our solution enables efficient data transfer between the diver and the buoy, which holds information that the developers can extract. This two-sided communication ensures real-time exchange of vital information underwater, enhancing the efficacy and safety of diving operations.

### *A. MATLAB*

The MATLAB application is an interface where the developers can send data to the diver in various forms, such as a text message, frequency (Hz) pulse width (for piezo tone buzzer), and an audio file (.wav file). This application also serves as a platform for developers to monitor the diver's feedback responses, manipulate the volume of the speaker, as well as log the data/messages sent to the diver.

A key component of this application is the "text-to-speech" feature, where the user can type a string value into a text field and click the "send message" button to synthesize the speech signal using an IBM pre-trained text-to-speech model. This is possible with the "speechClient()" and "text2speech" MATLAB API, which takes the input of the given text and pre-trained model and outputs the speech and its generated resonant frequency. This feature allows for communication between the developer and diver in a clear, concise, and convenient way that further allows the diver and developer to exchange valuable information or data.

Another important component of the application is Bluetooth, which the application uses to transmit and receive data to and from the diver via the buoy. The current configuration labels the MATLAB application as the central device that searches for an advertised Bluetooth connection, while the Arduino Nano ESP32 is labeled the peripheral device, as it sends the advertised Bluetooth connection and waits for the MATLAB application (central device) to secure the connection. Once the central and peripheral devices are connected, the MATLAB application notifies the developers of a secure connection by displaying a green light via The "Lamp" object on MATLAB, which is initially dim red (in color), which would indicate that the Bluetooth connection has not been made yet. When the user enters and sends a specific frequency and pulse width value using their respective text fields or enters a string of text to be converted to speech, our code stores this data and sends it to the speaker using MATLAB's Bluetooth object. Using the Bluetooth() MATLAB API, we created a Bluetooth object that connects to and represents the board. This object holds data through two characteristics (one contains data for a counter to communicate which rating the diver chooses to send; the other is a confirmation value that indicates when the diver confirms the rating they sent via the glove), which hold a Universally Unique Identifier (UUID), a 128-bit label/address used to identify each characteristic. This object can both have the data of the characteristics written to and read by it concerning the number of bytes available by using the pre-existing read() and write() MATLAB API for Bluetooth objects.

The application also serves as a means to log and monitor the data it sends and receives. To store the data received from the diver we created arrays of specific values and arranged them in data matrices. These matrices are then uploaded to a table to display each response sent to the diver. The interface also includes a meter to represent the diver's feedback on their current state, which changes by taking the feedback from the diver as input via Bluetooth and visually outputting this data to the developers. A key component of storing data sent by the divers is the Bluetooth characteristics, which the application can save as variables to then be able to access the real-time rating that the diver confirms and sends to the MATLAB application using Bluetooth.

### *C. Arduino*

Alongside the MATLAB application, we also have the Arduino Nano ESP32 integrated code. This code acts as a middle man allowing developers to send specific frequency ratings to the speaker, and receive data from the diver. The code works by counting the number of times the force sensor is held down, allowing for a count variable to increment by one. To keep the count within the bounds of five (the allocated rating values), we took the modulo of each number by 6. Once the diver has confirmed their reading by clicking the confirm force sensor, the code plays a tone on the watch and sends the values to the MATLAB app via Bluetooth. Along with this code, we are using Bluetooth characteristics that allow communication to and from the MATLAB device. By creating a characteristic for both confirm and count variables we can track and send values to MATLAB, making it easier to store and run statistical analysis on frequency data.

The speaker is connected to the *"SpeakerJack"* pin. The *"setup()"* function initializes the serial communication at a baud rate of 9600, which is used for data transmission between the microcontroller and the connected devices. In the *"loop()"* function, the code continuously checks if there is available data from the Bluetooth module using *"SerialBT.available()"*. If there is data, it reads the data using *"SerialBT.read()"* and assigns it to *"toneHz"*, which represents the frequency of the tone to be played on the speaker. The *"tone(SpeakerJack, toneHz, length(toneHzs))"* function generates a square wave of the specified frequency (and 50% duty cycle) on the *"SpeakerJack"* pin, causing the speaker to produce a sound. The code then reads the values from the two sensor pins using "*analogRead()"*.

In conclusion, the ESP32 Nano proved to be a highly advantageous choice for developing this system, offering a range of benefits that significantly enhanced its functionality. With its WiFi and Bluetooth capabilities enabling seamless wireless communication, efficient power management, and adaptability, researchers could tailor the device to meet specific research requirements. Overall, the ESP32 stood out as the optimal microcontroller due to its cost-effectiveness, efficiency, rapid development capabilities, and abundant resources, making it the ideal choice for creating this system.

## IV. DATA ANALYSIS & DISCUSSION

To check the efficiency of our device and the acoustic capabilities of some frequencies, we designed a study that would allow us to test different frequencies underwater. Due to the lack of a large body of water or pool, we used a shallow bucket of water and safety ear muffs to mimic the conditions under the ocean. Our criteria included finding playing different tones under shallow water, listening for specific frequencies, and assigning a rating to each of these frequencies to run a statistical test. To lower bias in our dataset, we had six different participants measuring five different frequencies (values between 0-255). This allowed us to run ANOVA's single factors test with an Honest Significant Difference test (HSD test) to determine significant differences in how people perceive sound underwater (shown in Figure 6).

Figure 6

| Frequency (HZ) | User 1 | User 2 | User 3 | User 4 | User 5 | User 6 | |
|---|---|---|---|---|---|---|---|
| 50 | 2 | 3 | 0 | 2 | 1 | 0 | 1.333333333 |
| 100 | 5 | 2 | 3 | 4 | 2 | 4 | 3.333333333 |
| 150 | 4 | 3 | 5 | 2 | 3 | 5 | 3.666666667 |
| 200 | 3 | 4 | 3 | 4 | 3 | 5 | 3.666666667 |
| 255 | 4 | 4 | 5 | 5 | 5 | 5 | 4.666666667 |

Anova: Single Factor

SUMMARY

| Groups | Count | Sum | Average | Variance |
|---|---|---|---|---|
| Row 1 | 6 | 8 | 1.333333333 | 1.466666667 |
| Row 2 | 6 | 20 | 3.333333333 | 1.466666667 |
| Row 3 | 6 | 22 | 3.666666667 | 1.466666667 |
| Row 4 | 6 | 22 | 3.666666667 | 0.666666667 |
| Row 5 | 6 | 28 | 4.666666667 | 0.266666667 |

ANOVA

| Source of Variation | SS | df | MS | F | P-value | F crit |
|---|---|---|---|---|---|---|
| Between Groups | 36 | 4 | 9 | 8.4375 | [illegible] | 2.758710487 |
| Within Groups | 26.66666667 | 25 | 1.066666667 | | | |
| | | | | | | |
| Total | 62.66666667 | 29 | | | | |

| Honest Significance Difference (HSD) Test | Group Diff. | 1.397862147 |
|---|---|---|
| 50Hz - 100Hz | 2.0000 | TRUE |
| 50Hz - 150Hz | 14.0000 | TRUE |
| 50 - 200 hz | 14.0000 | TRUE |
| 50 - 255 | 20.0000 | TRUE |
| 100 - 150 | 2.0000 | TRUE |
| 100 - 200 | 2.0000 | TRUE |
| 100 - 255 | 8.0000 | TRUE |
| 150 - 200 | 0.0000 | FALSE |
| 150 - 255 | 6.0000 | TRUE |
| 200 - 255 | 6.0000 | TRUE |

Figure 6: statistical analysis of psychoacoustic water ratings

As seen above, this test shows that there was a significant difference between each frequency except 150 and 200. More importantly, it shows that there is an honest significant difference between different frequencies underwater and that people can perceive different frequencies similarly. By testing and getting more data we could potentially determine more statistically significant information regarding psychoacoustic perception of sound underwater for freed divers.

## V. FINDINGS & FUTURE WORK

Reflecting on the development of SonicSync, several key insights have emerged, shedding light on both current achievements and avenues for future exploration. One pivotal realization is the imperative integration of Bluetooth technology and a buoy system. Recognizing the limitations imposed by underwater conditions, where traditional wireless communication methods falter due to water's density and its absorption and attenuation of radio waves, we found that employing Bluetooth via a buoy offers a promising solution. This approach not only extends the range and efficacy of wireless connections but also maximizes battery potential, aligning with the research criteria outlined by Duke. Additionally, while acoustic communication proves effective, challenges persist, suggesting the potential for exploring new underwater backscatter techniques, albeit with inherent complexity [22].

Another significant finding pertains to the necessity of a dedicated power source for the speaker component. Addressing the high power dissipation and current requirements essential for breaking the resonance frequency of water, our experiments led to the adoption of a separate battery. Operating the ESP32 circuit at 12 Volts and powering the exciter with a 14.8V 7200mAh battery proved optimal for achieving the desired frequency range, up to 70Hz.

Furthermore, a crucial insight emerged concerning human interaction with the device. The propensity for accidental taps or movements, particularly evident during swimming activities, underscores the importance of developing error detection systems to prevent inadvertent adjustments or diver feedback disruptions.

In future endeavors, building upon these insights will be essential for refining the device's functionality, enhancing its reliability, and expanding its applicability in subaqueous acoustic research and related fields.

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## References

[1] J. Mabrouki, M. Azrour, D. Dhiba, Y. Farhaoui, and S. E. Hajjaji, "IoT-based Data Logger for Weather Monitoring Using Arduino-based Wireless Sensor Networks with Remote Graphical Application and Alerts," *Big Data Mining and Analytics*, vol. 4, no. 1, pp. 25–32, Mar. 2021, doi: https://doi.org/10.26599/bdma.2020.9020018.

[2] E. Morrison and NAVY EXPERIMENTAL DIVING UNIT PANAMA CITY FL, "Evaluation of Digital Diving Watches.," *apps.dtic.mil*, Jan. 01, 1982. https://apps.dtic.mil/sti/citations/ADA133465 (accessed Apr. 19, 2024).

[3] F. M. White, *Fluid Mechanics*. New Delhi, India: McGraw-Hill Education, 2020. Accessed: Apr. 19, 2024. [Online]. Available: http://ftp.demec.ufpr.br/disciplinas/TM240/Marchi/Bibliografia/White_2011_7ed_Fluid-Mechanics.pdf

[4] S. Nieukirk and Acoustic Monitoring Project NOAA Pacific Marine Environmental Laboratory, "Understanding Ocean Acoustics," *NOAA.gov*,2015. https://oceanexplorer.noaa.gov/explorations/sound01/background/acoustics/acoustics.html#:~:text=When%20underwater%20objects%20vibrate%2C%20they (accessed Apr. 19, 2024).

[5] W. Ubellacker, "Underwater Communication via Compact Mechanical Sound Generation," *dspace.mit.edu*, 2013. https://dspace.mit.edu/handle/1721.1/83751 (accessed Apr. 19, 2024).

[6] F. Ragozzine, "Using Audacity for Demonstrations of Psychoacoustical Principles," *Teaching of Psychology*, vol. 39, no. 4, pp. 252–261, Oct. 2012, doi: https://doi.org/10.1177/0098628312456616.

[7] W. Grussenmeyer, J. Garcia, and F. Jiang, "Feasibility of Using Haptic Directions through Maps with a Tablet and Smart Watch for People Who Are Blind and Visually Impaired," pp. 83–89, Sep. 2016, doi: https://doi.org/10.1145/2935334.2935367.

[8] M. Sivaguru, S. Ambusam, B. K., P. Vinosh Kumar, and V. Rajkumar Krishnan, "Assessment of Vo2 Max Reliability with Garmin Smart Watch among Swimmers," *International Journal of Pharma and Bio Sciences*, vol. 11, no. 4, pp. L42-46, Jul. 2021, doi: https://doi.org/10.22376/ijpbs/lpr.2021.11.4.l42-46.

[9] E. J. Allen, P. C. Burton, C. A. Olman, and A. J. Oxenham, "Representations of Pitch and imbre Variation in Human Auditory Cortex," *The Journal of Neuroscience*, vol. 37, no. 5, pp. 1284–1293, Dec. 2016, doi: https://doi.org/10.1523/jneurosci.2336-16.2016.

[10] T. Benson, "Speed of Sound," *www.grc.nasa.gov*, Jan. 01, 2019. https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/sound.html#:~:text=The%20speed%20of%20sound%20(a (accessed Apr. 19, 2024).

[11] W. Yuyan, G. Lin, and H. Xiangdong, "A Review of Research on the Neurocognition for Timbre Perception," *Frontiers in Psychology*, 2022. https://www.frontiersin.org/articles/10.3389/fpsyg.2022.869475 (accessed Apr. 19, 2024).

[12] A. J. Oxenham, "Pitch Perception," *Journal of Neuroscience*, vol. 32, no. 39, pp. 13335–13338, Sep. 2012, doi: https://doi.org/10.1523/jneurosci.3815-12.2012.

[13] J. H. McDermott and A. J. Oxenham, "Music perception, pitch, and the auditory system," *Current Opinion in Neurobiology*, vol. 18, no. 4, pp. 452–463, Aug. 2008, doi: https://doi.org/10.1016/j.conb.2008.09.005.

[14] S. E. Martin, "Protecting electrical and electronic systems from moisture damage," *OCEANS 2009*, Oct. 2009, doi: https://doi.org/10.23919/oceans.2009.5422477.

[15] B. Heisler, "Communication Architecture and Protocols for an Underwater Stray Diver Alert System," *atrium.lib.uoguelph.ca*, Feb. 01, 2013. https://atrium.lib.uoguelph.ca/items/61b66349-2823-4344-aeff-4d99149a30c2 (accessed Apr. 19, 2024).

[16] S. You, W. Han, H.-J. Jang, and G.-Y. Noh, "Changes in Psychoacoustic Recognition and Brain Activity by Types of Fire Alarm," *International Journal of Environmental Research and Public Health*, vol. 18, no. 2, p. 541, Jan. 2021, doi: https://doi.org/10.3390/ijerph18020541.

[17] Parker Hannafin, “Parker O-Ring Handbook,” *parker.com*, 2021. https://www.parker.com/content/dam/Parker-com/Literature/O-Ring-division-Literature/ORD-5700.pdf (accessed Apr. 19, 2024).

[18] Z. Qin, Y.-T. Wu, L. He, X. Gao, and S.-K. Lyu, “Empirical Research on the Friction Behavior of O-rings in Hydraulic Cylinders,” *PLOS ONE*, vol. 18, no. 1, pp. e0280815–e0280815, Jan. 2023, doi: https://doi.org/10.1371/journal.pone.0280815.

[19] M. Mariello *et al.*, “Reliability of Protective Coatings for Flexible Piezoelectric Transducers in Aqueous Environments,” *Micromachines*, vol. 10, no. 11, p. 739, Oct. 2019, doi: https://doi.org/10.3390/mi10110739.

[20] T. Shimura, S. Sato, P. Zalar, and N. Matsuhisa, “Engineering the Comfort‐of‐Wear for Next Generation Wearables,” *Advanced Electronic Materials*, vol. 9, no. 9, p. 2200512, Oct. 2022, doi: https://doi.org/10.1002/aelm.202200512.

[21] E. Staff, “Making Capacitive Touch Sensors Water Tolerant,” *Embedded.com*, Jul. 31, 2011. https://www.embedded.com/making-capacitive-touch-sensors-water-tolerant/ (accessed Apr. 19, 2024).

[22] J. Jang and F. Adib, “Underwater backscatter networking,” MIT Media Lab, Aug. 2019, doi: https://doi.org/10.1145/3341302.3342091.